CYBERSECURITY • AWARENESS SERIES

CNS ALA-3 : Cyber Deception Chronicles



ATTACKS & MAJOR SECURITY BREACHES

How attackers manipulate people — and how we can stop them.

PEOPLE • PROCESS • TECHNOLOGY

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Social engineering: attacks on human trust

Attackers exploit urgency, authority, curiosity, fear, or helpfulness—not just software weaknesses.

01 PHISHING

Fake emails, texts, or websites imitate trusted organisations to steal passwords, payment details, or one-time codes.

RED FLAGS

- Urgent account warning
- Unexpected attachment/link
- Look-alike sender address

02 PRETEXTING

An attacker invents a believable story—such as being IT support or a supplier—to persuade someone to reveal data or grant access.

RED FLAGS

- Unusual request from authority
- Pressure to bypass policy
- Identity cannot be verified

03 BAITING

A tempting offer or item—such as a “free” download or found USB—encourages a victim to install malware or expose information.

RED FLAGS

- Unknown USB/device
- Too-good-to-be-true offer
- Unapproved download

Twitter, 2020: trusted employees became the target

A high-profile example of social engineering enabling access to powerful internal tools.

HOW IT UNFOLDED

1. **Target employees** — Attackers used phone-based social engineering to target employees.
2. **Gain internal access** — Access to internal systems enabled account-control capabilities.
3. **Abuse trusted tools** — Compromised tools were used to take over prominent accounts.
4. **Public impact** — Fraudulent cryptocurrency-related posts appeared on well-known accounts.

WHAT WENT WRONG?

- Human trust was exploited.
- Internal tools had high impact.

HOW TO REDUCE THE RISK

- Verify identity independently.
- Limit admin access and monitor use.
- Train staff for social-engineering calls.

Case summary based on publicly reported incident details; exact attack paths can be complex.

From manipulation to breach

Use this flowchart to explain how a small human interaction can become a major security incident.



BREAK THE CHAIN

VERIFY

Confirm sender using a known channel

PROTECT

Use unique passwords + phishing-resistant MFA

REPORT

Notify security/IT immediately

Remember: attackers often rely on urgency. Pause, verify, then act.

Make the safe action the easy action

People, process, and technology should work together to prevent and contain attacks.

01 PAUSE & VERIFY

Be cautious with urgent requests. Confirm unexpected payment, password, or access requests through a known phone number or official channel.

02 SECURE ACCOUNTS

Use unique passwords with a password manager. Enable MFA; prefer passkeys or phishing-resistant security keys where available.

03 LIMIT ACCESS

Give users only the access they need. Protect administrator accounts and review access regularly.

04 HANDLE LINKS & DEVICES

Check domains before signing in. Avoid unknown attachments, downloads, QR codes, and USB devices.

05 REPORT QUICKLY

Report suspicious messages and calls. If you clicked or shared details, contact IT/security immediately and change affected credentials safely.

06 PRACTISE & PREPARE

Run awareness training and simulations. Keep software updated, back up important data, and rehearse incident response.

Trust carefully. Verify independently.

Social engineering succeeds when a message feels convincing enough to bypass normal checks.

THE 3-STEP HABIT

1. **STOP** — Do not react to pressure or urgency.
2. **CHECK** — Verify the person, link, and request.
3. **REPORT** — Tell IT/security about suspicious activity.

REFERENCES & FURTHER READING

- **FBI IC3 — Social Engineering / Cybersecurity PSA**
  https://www.ic3.gov/PSA/2024/PSA240411
- **CISA — Cybersecurity Advisories**
  https://www.cisa.gov/news-events/cybersecurity-advisories/aa23-059a
- **Twitter 2020 incident — public reporting**
  Search: Twitter 2020 social engineering breach official investigation

FINAL MESSAGE **Think before you click. Verify before you trust. Report before it spreads.**

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